

NREDI: physics-verified periorbital wearable with multimodal AI for joint diabetic retinopathy and neuropathy screening

This notebook reproduces every headline number of the project from the stored outputs. It reads `results/RESULTS_MASTER.json` (the single source of numbers) and the per-subject prediction files in `results/out/`, recomputes the metrics live with scikit-learn, and draws the plots. No raw images or patient rows are loaded here; retraining is done by the scripts in `results/src/` (see the README).

Datasets (all public or open): DeepDRiD (CC BY-SA 4.0), IDRiD (CC BY 4.0), APTOS 2019 (Kaggle), Istanbul e-Nabiz EHR export (Mendeley Data, CC BY 4.0), PhysioNet Novak-lab diabetes cohorts (CC BY 4.0), PPG-BP (figshare, CC0). Links are in the README.

```
In [1]: import json, pathlib, numpy as np, pandas as pd, matplotlib.pyplot as plt
from sklearn.metrics import roc_auc_score, roc_curve, cohen_kappa_score, confusion_
ROOT = pathlib.Path.cwd()
while not (ROOT / "results" / "RESULTS_MASTER.json").exists(): ROOT = ROOT.parent
OUT = ROOT / "results" / "out"; SIM = ROOT / "sim" / "out"
M = json.loads((ROOT / "results" / "RESULTS_MASTER.json").read_text(encoding="utf-8"))
def v(*ks):
    d = M
    for k in ks: d = d[k]
    return d["value"] if isinstance(d, dict) and "value" in d else d
def ci(*ks):
    d = M
    for k in ks: d = d[k]
    return d.get("ci95")
print("results master built", M["_meta"]["built"], "|", len(M), "sections")
```

results master built 2026-09-08T07:46:29 | 20 sections

1. Fundus retinopathy grader (ConvNeXt-Tiny, ordinal regression, patient-grouped 5-fold on DeepDRiD)

Out-of-fold predictions are in `fundus_oof.npz`; the four thresholds fitted on them are applied unchanged to IDRiD and APTOS 2019.

```
In [2]: # our own outputs (object arrays hold image paths), so allow_pickle is safe
Z = np.load(OUT / "fundus_oof.npz", allow_pickle=True); I = np.load(OUT / "fundus_i
th = np.array(v("fundus_dr", "thresholds")); m = ~np.isnan(Z["oof"])
grade = lambda s: np.digitize(s, np.sort(th))
rows = []
for name, y, s in (("DeepDRiD OOF (China)", Z["grade"][m], Z["oof"][m]), ("IDRiD ex
g = grade(s))
```

```
rows.append({"dataset": name, "n": len(y), "QWK": cohen_kappa_score(y, g, weigh
"sensitivity": ((g >= 2) & (y >= 2)).sum() / (y >= 2).sum(), "spec
df = pd.DataFrame(rows).set_index("dataset").round(3); df
```

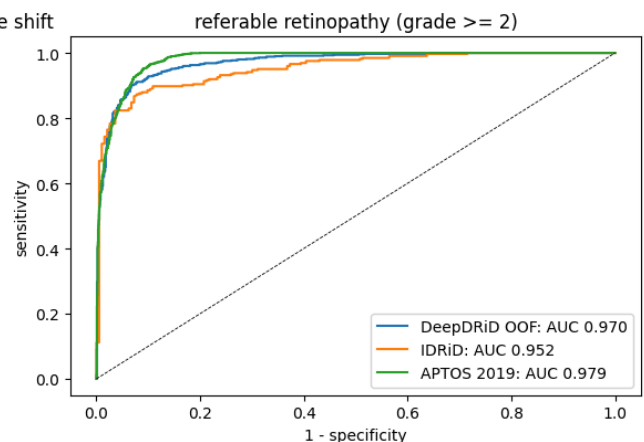
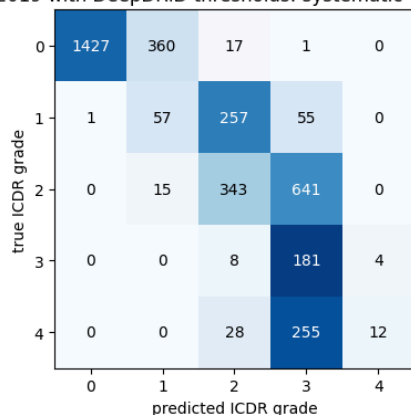
Out[2]:

	n	QWK	AUC referable	accuracy	sensitivity	specificity
dataset						
DeepDRiD OOF (China)	1589	0.857	0.970	0.685	0.901	0.933
IDRiD external (India)	516	0.729	0.952	0.450	0.938	0.715
APTOS 2019 external (India)	3662	0.845	0.979	0.552	0.990	0.848

```
In [3]: print("master says: OOF QWK", round(v("fundus_dr", "qwk"),3), ci("fundus_dr", "qwk"),
fig, ax = plt.subplots(1, 2, figsize=(11, 4.2))
cm = confusion_matrix(A["grade"], grade(A["pred"]), labels=range(5))
im = ax[0].imshow(cm / cm.sum(1, keepdims=True), cmap="Blues", vmin=0, vmax=1)
for i in range(5):
    for j in range(5): ax[0].text(j, i, cm[i, j], ha="center", va="center", color="
ax[0].set_xlabel("predicted ICDR grade"); ax[0].set_ylabel("true ICDR grade"); ax[0
for name, y, s in (("DeepDRiD OOF", Z["grade"][m], Z["oof"][m]), ("IDRiD", I["grade
f, t, _ = roc_curve(y >= 2, s); ax[1].plot(f, t, label=f"{name}: AUC {roc_auc_s
ax[1].plot([0, 1], [0, 1], "k--", lw=0.6); ax[1].set_xlabel("1 - specificity"); ax[
plt.tight_layout(); plt.show()
```

master says: OOF QWK 0.857 [0.8348696995260773, 0.8776324753621546] | APTOS QWK 0.845 | APTOS re-fitted thresholds (reported, not used): 0.878

APTOS 2019 with DeepDRiD thresholds: systematic +1 grade shift



2. Macular edema head (IDRiD, DME grade 0 to 2)

```
In [4]: D = np.load(OUT / "dme_idrid_pred.npz"); thd = np.array(v("dme", "thresholds"))
print("OOF (413 images): QWK", round(cohen_kappa_score(D["y_train"], np.digitize(D[
print("fixed test (103 images): QWK", round(cohen_kappa_score(D["y_test"], np.digit
ct = pd.crosstab(pd.Series(I["grade"], name="DR grade"), pd.Series(I["dme"], name="
```

OOF (413 images): QWK 0.916 | any-DME AUC 0.982

fixed test (103 images): QWK 0.838 | any-DME AUC 0.945

Out[4]: **DME grade** 0 1 2

DR grade				
0	168	0	0	
1	25	0	0	
2	16	36	116	
3	6	7	80	
4	7	8	47	

3. Systemic branch on routine records (Istanbul export, n = 77,724)

The neuropathy model **excludes treatment proxies** (antiepileptics = gabapentin/pregabalin, ATC A16 = alpha-lipoic acid, other nervous-system drugs). They follow the diagnosis rather than precede it. Keeping them inflates the AUC to 0.838; the honest number is 0.710.

```
In [5]: E = np.load(OUT / "ehr_oof.npz")
rows = [("retinopathy, logistic", E["y_dr"], E["retinopathy_logistic"]), ("retinopa
("neuropathy, logistic (proxies excluded)", E["y_dn"], E["diabetic_neuropat
("sensitivity: neuropathy WITH treatment proxies", E["y_dn"], E["DN_with_pr
pd.DataFrame([{"model": n, "n": len(y), "prevalence": y.mean().round(4), "AUC (5-f
```

Out[5]:

	n	prevalence	AUC (5-fold OOF)
--	---	------------	------------------

model			
retinopathy, logistic	77724	0.1399	0.710
retinopathy, boosted	77724	0.1399	0.720
neuropathy, logistic (proxies excluded)	77724	0.1734	0.703
neuropathy, boosted (proxies excluded) <- reported	77724	0.1734	0.710
sensitivity: neuropathy WITH treatment proxies	77724	0.1734	0.838
sensitivity: also without analgesics/psychoanaleptics	77724	0.1734	0.708

```
In [6]: jo = M["ehr"]["joint_ovr_auc"]["value"]; cnt = M["ehr"]["joint_counts"]["value"]
print("joint 4-class one-vs-rest AUC:", {k: round(x, 2) for k, x in jo.items()})
print("DR-DN co-occurrence odds ratio", round(v("ehr", "dr_dn_odds_ratio"), 2), "Fi
print("top logistic odds ratios per SD (neuropathy):", {k: round(x, 2) for k, x in
```

joint 4-class one-vs-rest AUC: {'none': 0.72, 'DR only': 0.68, 'DN only': 0.68, 'both': 0.78} | counts: {'0': 56255, '1': 7994, '2': 10597, '3': 2878}
 DR-DN co-occurrence odds ratio 1.91 Fisher p 1.32180626434428e-147
 top logistic odds ratios per SD (neuropathy): {'HbA1c': 1.2, 'age': 1.18, 'musculoskeletal_dis': 1.18, 'insulin_aspart': 1.17, 'lipoprotein_met_dis': 1.16}

4. Physiological branch: heart-rate variability (PhysioNet) and fingertip PPG (PPG-BP)

One positive result and two honest nulls, each with a positive control.

```
In [7]: H = np.load(OUT / "hrv_pred.npz"); P = np.load(OUT / "ppg_pred.npz")
print("HRV: diabetes vs control LOO AUC", round(roc_auc_score(H["dm_vs_control_y"],
print("HRV: retinopathy within diabetics LOO AUC", round(roc_auc_score(H["dr_within
print("HRV: autonomic symptoms AUC", round(roc_auc_score(H["autonomic_symptoms_y"],
for k in ("dm_demographics_only", "dm_morphology", "dm_morphology_plus_demo", "dm_p
    print(f"PPG-BP {k:26s} AUC {roc_auc_score(P[k+'_y'], P[k+'_p']):.3f}")
print("PaPaGei age control: r =", round(np.corrcoef(P["age"], P["age_pred"])[0, 1],
```

HRV: diabetes vs control LOO AUC 0.777 (n = 94)
 HRV: retinopathy within diabetics LOO AUC 0.367 (n = 58, NULL)
 HRV: autonomic symptoms AUC 0.541
 PPG-BP dm_demographics_only AUC 0.572
 PPG-BP dm_morphology AUC 0.372
 PPG-BP dm_morphology_plus_demo AUC 0.417
 PPG-BP dm_papagei_pca16 AUC 0.451
 PPG-BP dm_papagei_plus_demo AUC 0.490
 PPG-BP htn_morphology AUC 0.726
 PaPaGei age control: r = 0.38 MAE 13.9 years

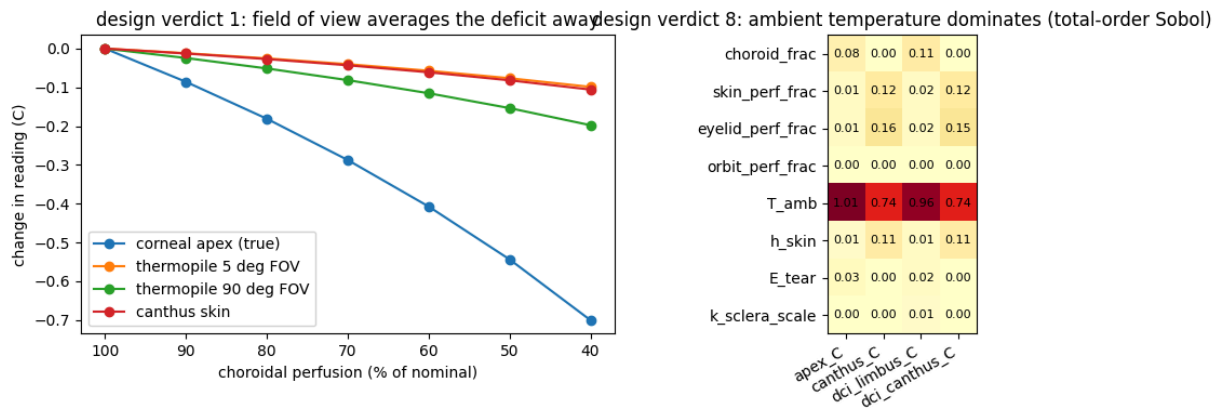
5. Digital twin: thermal forward model, verification and uncertainty

Numbers come from `sim/out/thermal3d_full.json`, `verification.json` and `surrogate_thermal.json` through the master.

```
In [8]: T = M["thermal"]; sw = T["choroid_sweep"]["value"]; base = sw[0]
print("apex %.2f C, canthus %.2f C, temple %.2f C at 25 C ambient; published apex %
print("MMS observed order %.2f; max GCI %.2f %%; apex deficit at 50 %% choroidal pe
fig, ax = plt.subplots(1, 2, figsize=(11, 4))
cf = [r["choroid_frac"] * 100 for r in sw]
for k, lab in (("apex_C", "corneal apex (true)"), ("dci_C", "thermopile 5 deg FOV"))
    ax[0].plot(cf, [r[k] - base[k] for r in sw], "o-", label=lab)
ax[0].invert_xaxis(); ax[0].set_xlabel("choroidal perfusion (% of nominal)"); ax[0]
ST = np.array(M["uq"]["ST"]["value"]); params = [p[0] for p in M["uq"]["params_rang
im = ax[1].imshow(np.clip(ST, 0, 1), cmap="YlOrRd", vmin=0, vmax=1); ax[1].set_xtic
for i in range(ST.shape[0]):
    for j in range(ST.shape[1]): ax[1].text(j, i, f"{ST[i,j]:.2f}", ha="center", va
ax[1].set_title("design verdict 8: ambient temperature dominates (total-order Sobol
print("apex slope %.2f C per C ambient (R2 %.2f)" % (v("uq", "apex_slope_per_C_ambie
```

apex 32.96 C, canthus 33.91 C, temple 34.47 C at 25 C ambient; published apex [34.0, 35.0], canthus [35.0, 36.5]

MMS observed order 1.79; max GCI 2.00 %; apex deficit at 50 % choroidal perfusion - 0.54 C; thermopile 90 deg captures 28 %, 5 deg 14 %



apex slope 0.32 C per C ambient (R2 0.84)

6. Optical, electrical and pupil forward models

```
In [9]: O = M["optical"]; B = M["bioimpedance"]; Pp = M["pupil"]
print("PPG at 3 mm: arterial share of attenuation 660 nm %.1f %, 880 nm %.1f %"; a
print("940 nm illuminator: IEC 62471 limit %.1f mW; heating at that power apex %.2f
print("bioimpedance at 50 kHz: orbit+eye+lid share %.1f %; contact share of a bipo
ss = Pp["stimulus_sweep"]["value"]; print("pupil: amplitude at 0 lx %.2f mm (940 nm
```

PPG at 3 mm: arterial share of attenuation 660 nm 0.7 %, 880 nm 2.1 %; at 12 mm the arterial share of the pulse is 35 % but photons fall 37.5-fold

940 nm illuminator: IEC 62471 limit 42.3 mW; heating at that power apex 0.21 C, lid 0.14 C, canthus 0.06 C

bioimpedance at 50 kHz: orbit+eye+lid share 15.6 %; contact share of a bipolar reading 68 %; edema +10 % water -> tetrapolar -1.39 % vs bipolar total -0.13 %

pupil: amplitude at 0 lx 0.10 mm (940 nm only) vs 2.23 mm at 300 lx; camera under-read 36.9 %

7. The eight design verdicts

Each verdict names the master entries behind it (see `results/RESULTS_MASTER.json` -> `design_verdicts`).

```
In [10]: for k, val in M["design_verdicts"].items(): print(k, "->", val if isinstance(val, l
```

1_thermopile_90deg_fov_too_wide -> ['thermal.thermopile_90deg_spot_mm', 'thermal.thermopile_90deg_capture_fraction', 'thermal.thermopile_5deg_capture_fraction']
2_940nm_illuminator_evokes_no_reflex -> ['pupil.stimulus_sweep']
3_aim_thermopile_at_ocular_surface_not_canthus -> ['uq.ST_choroid_ocular_readouts', 'uq.S1_choroid_per_output']
4_electrode_contact_model -> ['bioimpedance.electrodes', 'bioimpedance.bipolar_contact_share_of_total_50kHz']
5_3mm_ppg_reads_dermal_plexus_not_artery -> ['optical.ppg_by_wavelength', 'optical.spacing_sweep_880', 'optical.detected_fraction_ratio_3mm_over_12mm']
6_cross_eye_tetrapolar_blind_to_orbit -> ['bioimpedance.orbit_share_50kHz', 'bioimpedance.edema_50kHz']
7_illuminator_heating_confound -> ['opto_thermal.apex_rise_at_iec_limit_C', 'opto_thermal.apex_rise_as_fraction_of_50pct_choroid_deficit']
8_ambient_temperature_dominates_compensate -> ['uq.ST_ambient_range', 'uq.apex_slope_per_C_ambient']

Integrity notes

- No quantum layer contributed to any result (`results/src/concept_quantum.py` was smoke-tested only and produced no output).
- The neuropathy AUC reported is 0.710 with treatment proxies excluded; 0.838 is a sensitivity analysis only.
- APTOS: ordering transfers (AUC 0.979) but grade boundaries do not (accuracy 55 %); re-fitted thresholds (QWK 0.878) are reported, not used.
- The wearable is a verified design; no prototype has been tested on people and no human data were generated.